9th Class • Punjab Board • PECTAA New Syllabus

Chapter 8: Ethical, Social, and Legal Concerns in Computer Usage

Complete solved exercise — safety, privacy, responsible behaviour, data ethics and software piracy.

Chapter 8 • Complete official exercise • 9 MCQs + 8 short + 3 long questions

What this chapter is about

- **Safe computing** protects accounts, devices and data through updates, strong authentication, careful downloads, secure networks and backups.
- **Responsible computing** also considers honesty, respect, privacy, intellectual property and the effect of online behaviour on other people.
- **Data ethics** asks whether data collection and use are fair, necessary, transparent, secure and respectful of people's rights—not merely whether they are technically possible.
- Privacy laws and software licences create legal duties. Ethical behaviour may require more than the legal minimum, especially when information is sensitive or people are vulnerable.

Coverage: Chapter 8 corresponds to Unit 11 Ethical, Social, and Legal Concerns in Computer Usage in the twelve-unit textbook. The April 2026 nine-unit exercise is solved in book order.

A. Multiple Choice Questions

1. Why is it important to use computers safely and responsibly?

(a) To ensure we can use them more frequently

(b) To protect our personal information and make wise choices about hardware and software

(c) To make the computer run faster

(d) To avoid paying for software

Answer: (b) To protect our personal information and make wise choices about hardware and software — Responsible use protects information, devices and people and includes suitable, compatible technology choices.

2. What does “responsible computer usage” include?

(a) Choosing the most expensive hardware

(b) Sharing your passwords with friends

(c) Being cautious about what you share online and protecting yourself and others

(d) Ignoring software updates

Answer: (c) Being cautious about what you share online and protecting yourself and others — Responsible users consider privacy, safety and the effect of their online actions on other people.

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3. What should you check to ensure hardware and software compatibility?

(a) The color of the hardware

(b) The system requirements on software packages and match them with your computer's specifications

(c) The price of the hardware

(d) The brand of the hardware

Answer: (b) The system requirements on software packages and match them with your computer's specifications — Match the required operating system, processor, memory, storage and other specifications with the computer before installation or purchase.

4. Why is it important to use strong, unique passwords?

(a) To make your accounts easy to hack

(b) To help others guess your password

(c) To make it harder for someone to guess your password and access your accounts

(d) To avoid installing software updates

Answer: (c) To make it harder for someone to guess your password and access your accounts — Length and uniqueness make guessing and reuse attacks harder, so one leaked password does not open every account.

5. What is one reason to avoid clicking on unknown links or downloading files from untrusted sources?

(a) They might contain helpful software

(b) They might contain malware that can harm your device or steal your information

(c) They are usually cheaper

(d) They help your device run faster

Answer: (b) They might contain malware that can harm your device or steal your information — Untrusted links and downloads can lead to phishing, malicious websites or malware installation.

6. What does Two-Factor Authentication (2FA) do?

(a) It makes your password easier to guess

(b) It adds an extra layer of security by requiring a second form of verification

(c) It removes the need for a password

(d) It reduces the need for software updates

Answer: (b) It adds an extra layer of security by requiring a second form of verification — 2FA requires another independent proof, so a stolen password alone is normally insufficient.

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7. Why should you be cautious when using public Wi-Fi for sensitive transactions?

(a) Public Wi-Fi is usually faster
(b) Pubtic Wi-Fi networks are often less secure
(c) Public Wi-Fi is free
(d) Public Wi-Fi always provides encryption

Answer: (b) Pubtic Wi-Fi networks are often less secure — Public networks may be poorly controlled or impersonated, exposing users to interception or malicious hotspots.

8. What is an important aspect of responsible use of social media?

(a) Sharing personal information like your home address
(b) Posting photos without considering privacy settings
(c) Being respectful and avoiding sharing sensitive information publicly
(d) Ignoring privacy settings

Answer: (c) Being respectful and avoiding sharing sensitive information publicly — Respectful behaviour and careful control of sensitive information protect the user and others.

9. What should you do if you receive an email from an unknown sender asking for personal information?

(a) Provide the information requested
(b) Forward the email to your friends
(c) Ignore or delete the email
(d) Open the email and click on any links

Answer: (c) Ignore or delete the email — Do not provide information or open its links; delete or report the suspicious message through a trusted route.

B. Short Questions

1. What is the importance of using computers safely and responsibly?

Safe and responsible use protects personal information, accounts, hardware and software from harm. It also prevents legal and ethical misuse, supports reliable work and protects other people from fraud, harassment and privacy violations.

2. Why should you create strong, unique passwords for your accounts?

A strong password is difficult to guess, and a unique password prevents one leaked account password from unlocking other accounts. Use long passphrases, avoid personal details and never reuse or share passwords; a password manager can help.

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3. How can you protect yourself from harmful links and downloads?

Do not open unexpected links or attachments. Check the sender and full web address through a trusted route, download only from official sources, keep security software and the system updated, and scan files before opening. Delete and report suspicious requests rather than replying with information.

4. What is Two-Factor Authentication (2FA), and why is it useful?

2FA requires two different proofs of identity, such as a password plus a code from an authenticator app or a security key. It adds a second barrier, so an attacker who learns the password still cannot normally sign in.

5. Why is it a good idea to avoid using public Wi-Fi for sensitive transactions?

Public Wi-Fi may be insecure, monitored or imitated by a fake hotspot. Sensitive logins and payments are safer on a trusted private connection; if use is unavoidable, verify the network and HTTPS, avoid sensitive activity where possible and keep the device updated.

6. Why is it important to be careful about what you share on social media?

Posts can be copied, misunderstood or redistributed beyond the intended audience. Avoid sharing addresses, passwords, identity documents, live location and other sensitive details; use privacy settings, ask before posting others and remain respectful.

7. What is the purpose of privacy laws regarding your personal information?

Privacy laws set rules for collecting, using, storing, sharing and protecting personal information. They give people recognised rights and place duties on organisations, helping prevent unauthorised access, misuse and disclosure.

8. What is software piracy, and why is it harmful?

Software piracy is copying, distributing, installing or using software in violation of its licence or copyright. It harms developers and legitimate businesses, can lead to legal penalties and often exposes users to malware, missing updates and unreliable software.

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C. Long Questions

1. Discuss the importance of responsible computer usage in today's digital world.

Computers now store school records, photographs, conversations, financial details and work documents. A careless action can therefore affect privacy, money, reputation, safety and other people as well as the device itself.

• **Personal security:** Strong authentication, updates and cautious downloading reduce account theft, fraud and malware.

• **Privacy:** Careful sharing and suitable settings prevent sensitive information from reaching unintended audiences.

• **Reliable decisions:** Checking hardware/software requirements and file sources prevents incompatibility, data loss and unnecessary cost.

• **Respect and wellbeing:** Responsible users avoid cyberbullying, impersonation and harmful rumours and consider screen time and the effect of their words.

• **Legal and ethical duties:** Respecting copyright, licences, consent and privacy rules protects creators and the rights of others.

• **Academic and professional trust:** Honest work, correct attribution and protection of shared records make digital collaboration dependable.

• **Community safety:** Reporting suspicious messages and helping others use safe practices reduces the spread of scams and harmful content.

Responsible use is therefore more than avoiding viruses. It is the daily practice of being secure, truthful, respectful, lawful and thoughtful about long-term digital consequences.

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2. Describe the steps you should take to ensure the safe operation of digital platforms and devices.

Figure 1 — Safe operation is a cycle, not a one-time setup



Area	Actions	Reason
System and applications	Install trusted updates; remove unused software; use supported versions.	Patches close vulnerabilities and improve stability.
Accounts	Use unique passphrases, a password manager, 2FA and least-privilege permissions.	Limits guessing, reuse attacks and damage if one account is compromised.
Hardware/software compatibility	Compare system requirements, operating system, memory, storage, ports and drivers before installation.	Prevents crashes, data loss and unsuitable purchases.
Links and downloads	Verify senders and URLs independently; use official sources; scan files; refuse unexpected permissions.	Reduces phishing, malware and spyware.
Network use	Secure home Wi-Fi, change router defaults and avoid sensitive transactions on untrusted public networks.	Protects communication and connected devices.
Data protection	Make regular offline/off-site backups, encrypt sensitive information and test restoration.	Allows recovery after loss, theft, failure or ransomware.
Physical safety	Keep devices ventilated, dry and away from damaged cables; lock them when unattended.	Prevents overheating, electric hazards, theft and unauthorised use.
Online conduct	Use privacy settings, think before posting, respect others and report/block abuse.	Protects reputation, privacy and wellbeing.

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Area	Actions	Reason
Monitoring and response	Review account activity, read warnings and report incidents; change compromised credentials from a clean device.	Early action limits damage and helps prevent recurrence.

No single tool is sufficient. Safety comes from layered habits, regular review and knowing how to recover when something goes wrong.

3. Explain the concept of data ethics and its importance in handling personal and sensitive information.

Data ethics is the responsible evaluation of how data are collected, interpreted, used, shared, stored and discarded. It asks whether an action is fair and respectful to the people behind the data, even when technology makes the action easy.

Principle	Required practice
Purpose and necessity	Define a legitimate purpose and collect only data actually needed for it; do not reuse them secretly for an unrelated purpose.
Consent and transparency	Explain what is collected, why, who can access it and how long it is kept; obtain valid consent where required.
Data minimisation	Avoid unnecessary identity, contact, location or biometric details and reduce the number of copies.
Accuracy and fairness	Check records, correct mistakes and test whether decisions or AI models disadvantage particular groups.
Security and confidentiality	Use access control, encryption, secure devices and backups; share only with authorised people.
Retention and deletion	Keep information only as long as justified, then delete or anonymise it securely.
Rights and accountability	Respect requests for access/correction where applicable, record decisions and assign responsibility for mistakes or breaches.
Human oversight	Do not make harmful or high-stakes decisions solely from an unchecked score, profile or automated output.

For example, a school survey about study habits normally does not need students' home addresses. It should explain its purpose, collect the minimum answers, restrict access, report grouped results and delete identifiable responses when no longer needed. Ethical handling builds trust, reduces harm and supports legal compliance.

D. More Important Questions (from the text)

1. How can you recognise and respond to a phishing message?

• Warning signs include urgency, threats, an unexpected prize, an unusual sender address, generic greeting, spelling mistakes and requests for passwords, codes or payments.

• Do not click its link, open its attachment, reply with information or scan an unexpected code.

• Verify the request through an independently known app, phone number or website—not through contact details in the message.

• Report or delete it using the platform's tools. If information was already entered, change the affected and any reused passwords from a clean device, revoke sessions, enable 2FA and tell the relevant provider or responsible adult.

2. Differentiate between legitimate software use and software piracy.

Legitimate use	Software piracy
Follows the copyright licence and permitted number of users/devices.	Copies, shares, activates or installs outside the licence.
Uses official installers, valid licences or legitimately free/open-source terms.	Uses cracked activation, stolen keys or unauthorised copies; open-source also has licence conditions.
Receives proper updates, support and documentation where provided.	May lack updates/support and can contain hidden malware.
Respects developers' work and reduces legal/security risk.	Can cause legal penalties, financial harm and unreliable systems.

3. What should a student do about cyberbullying or harmful online content?

Do not respond with more abuse or forward the harmful material. Save appropriate evidence such as timestamps and URLs without unnecessarily spreading images, block the account and report it through the platform.

Tell a trusted adult, teacher or safeguarding contact, especially when threats, impersonation, private images or personal information are involved. Support the targeted person privately and avoid blaming them. Immediate danger or serious crime should be reported through the appropriate local authorities or emergency support.

Points to Remember

Layered security: updates + unique passphrases + 2FA + careful links/downloads + backups + safe networks.

2FA: two different proof factors; a second password is not the same as a second factor.

Public Wi-Fi: may be insecure or impersonated; avoid sensitive transactions on untrusted networks.

Privacy: collect/share less, use suitable settings and remember that posts can be copied permanently.

Data ethics: necessity, transparency, consent, minimisation, accuracy, fairness, security, retention and accountability.

Software piracy: licence/copyright violation with legal, security and reliability risks.

Phishing: verify independently; never send passwords or one-time codes to an unexpected requester.

Responsible response: preserve evidence safely, block/report, seek trusted help and change compromised credentials.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	b	2	c	3	b	4	c	5	b
6	b	7	b	8	c	9	c		

How to attempt this chapter in the paper

• For safety answers, organise points under accounts, devices, downloads, networks, data and online conduct instead of repeating passwords only.
• For data ethics, use a school or business example and explain both the principle and the action taken.
• For legal questions, distinguish a software licence from ownership of the physical/downloaded copy and avoid claiming that all free software has no conditions.
• For social scenarios, include protection of other people, evidence/reporting and recovery—not only personal device security.

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